Eric Petersen

Engineering Student at the University of British Columbia

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EDUCATION

University of British Columbia
Sep 2021 – Present

- Pursuing degree in Applied Science, in the Engineering Physics program
- 90.8% average, Dean's Honour List

AWARDS & CERTIFICATION

University of British Columbia, UTP Early Entrance Award
Sep 2021

- Awarded for early entrance to university, academic merit, leadership, and extracurricular involvement

Government of BC, BC Achievement Scholarship
Sep 2021

- Awarded to 8000 high school graduates with the top academic achievement in British Columbia

Dassault Systèmes, Certificate of Mechanical Design
Dec 2021

- Certified SolidWorks Associate (**CSWA**) in Mechanical Design

EXPERIENCE

UBC Aerodesign, Mechanical Engineer, Advanced Airfoils Team
Sep 2021 – Present

- Developed design constraints using **MATLAB** according to our aircraft mission profile
- Responsible for airfoil and 3D wing analysis using **XFLR**
- Sized structural components for wing loading in flight and landing impact
- Modelled internal vertical tail and rudder structure in **SolidWorks** using **DFM/A** principles
- Used **composite layups** and **laser cutting** to manufacture parts, from **AutoCAD** designs

BC Children's Hospital, Volunteer
July 2020

- Designed a medical device enclosure for **3D printing**, currently being used at the BC Children's Hospital
- Assisted with assembly of pediatric coping tools

TECHNICAL PROJECTS

Terrain Mapping Aircraft (Personal project)
July 2022 – Present

- Designing a low-speed RC airplane to map terrain accurately
- Evaluated flight stability and stall characteristics using **XFLR** and **SolidWorks Flow**

Autonomous Claw (UBC – APSC 101)
Jan 2022

- Developed code in **C** for ultrasonic sensor configuration to accurately detect distances within 1cm
- Employed a median filter in sensor readings to clean data outliers

Simon Game (UBC – APSC 160)
Nov 2021

- Programmed **Arduino** to play "Simon Says" with the user
- Used coding best practices including commenting, indentation, creating functions, and using local variables

TECHNICAL SKILLS

Programs: SolidWorks (CSWA) · AutoCAD · MATLAB · XFLR · SolidWorks Flow · SolidWorks PDM · MS Office
Languages: Java · C · JavaScript · HTML · CSS
Manufacturing: Laser Cutting · 3D Printing · Composite Layups